Mario Gomez Andreu

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Education

ETH Zürich, M.Sc. Robotics, Systems and Control — 09/2023 – Ongoing
Technical University of Darmstadt, B.Sc. Computer Science — 09/2020 – 08/2023

Research Experience

Trajectory Planning on 3D Gaussian Splats — 03/2024 – 03/2025
Robotic Systems Lab (RSL), ETH Zürich, [1]

- Developed **FOCI**, an algorithm for trajectory optimization on 3D Gaussian Splatting (3DGS) maps.
- Designed and implemented **GPU-accelerated collision computation** based on overlap integrals.
- Validated on the **ANYmal quadruped**.

Modelling for Universal Soft Lasso Gripper — 04/2024 – 09/2024
Robotic Systems Lab (RSL), ETH Zürich, [2]

- Co-authored paper on **rope-based robotic manipulation**.
- Contributed the **simulation for manipulator dynamics and contact interactions**.
- Evaluated fidelity against physical trials under quasi-static/contact conditions.

Optimization Based Motion Planning for Robotic Juggling — 04/2023 – 08/2023
Intelligent Autonomous Systems (IAS) Lab, TU Darmstadt, [3]

- First author of paper extending **robotic juggling from uniform to arbitrary siteswap sequences**.
- Developed a **bi-level planning framework** combining ball prediction with motion optimization.
- Demonstrated stability for all siteswap sequences (up to 9 balls) in simulation.

Robotic Tactile Exploration — 09/2022 – 03/2023
Intelligent Autonomous Systems (IAS) Lab, TU Darmstadt, [4]

- Contributed to an **active sampling approach for hardness classification** using vision-based tactile sensors.

Work Experience

Gravis Robotics AG, Zurich — 09/2024 – 03/2025
Internship

- Developed a **delay-aware MPC strategy** to improve latency handling in excavator control, achieving up to 20% faster in-air motions compared to the baseline;
- Designed a **collision-aware trajectory planner**.
- Validated in simulation and on hardware.

HS-Analysis GmbH, Karlsruhe — 05/2022 – 11/2022
Working Student

- Built a software module for **automated analysis of lateral flow assays** (e.g. COVID-19 tests).
- Integrated the module into the company's product and handled client requirements.

University Clinic, Hamburg-Eppendorf — 04/2021 – 03/2022
Research Assistant

- Designed a tool to convert structured data into graph representations (Neo4j).

German Cancer Research Center (DKFZ), Heidelberg — 07/2020 – 03/2021
Research Assistant

- Developed dashboards for visualization of medical data.

Leadership & Mentorship

Founding Board Member (Education), ETH Robotics Club 07/2025 – Ongoing

- Built club structure and organization, managed industrial and academic partnerships.
- Planned **10 talks and 5 workshops** scheduled for the club's first semester.

Undergraduate Thesis Co-supervisor, Universidad Pontificia Comillas (Spain) 04/2025 – Ongoing

- Provided guidance for two theses on **3DGS-based trajectory optimization** and **end-to-end learning in autonomous drone flight**.

Awards & Grants

German Academic Scholarship Foundation 06/2021 – Ongoing
Scholarship holder
IEEE IES-SYPA Travel Grant (IROS 2025) 2025
IEEE RAS Travel Support (IROS 2024) 2024

Academic Service

International Conference on Intelligent Robots and Systems (IROS) 2024, 2025
Reviewer

Skills

Programming: Python, C++, MATLAB
Libraries / Frameworks: PyTorch, CasADi, IPOPT, ROS/ROS2
Tools / DevOps: Git, Docker
Simulation: Gazebo, MuJoCo, IsaacGym
Languages: German (native), English (fluent), Spanish (intermediate)

Publications

[1] **Mario Gomez Andreu***, Maximum Wilder-Smith*, Victor Klemm, Vaishakh Patil, Jesus Tordesillas, and Marco Hutter. "FOCI: Trajectory Optimization on Gaussian Splats". In: *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025, pp. 1884–1891.

[2] Christian Friedrich, **Mario Gomez Andreu**, Gabriel Métois, Fan Shi, Marco Hutter, and Robert Baines. "RoboWrangler: Toward Rope-based Grasping for Mobile Manipulation". In: *IEEE International Conference on Soft Robotics (RoboSoft)*. IEEE, 2025.

[3] **Mario Gomez Andreu**, Kai Ploeger, and Jan Peters. "Beyond the Cascade: Juggling Vanilla Siteswap Patterns". In: *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2024, pp. 2928–2934.

[4] J. Chen, A. Kshirsagar, F. Heller, **Mario Gomez Andreu**, B. Belousov, T. Schneider, L. P. Y. Lin, K. Doerschner, K. Drewing, and J. Peters. "Active Sampling for Hardness Classification with Vision-Based Tactile Sensors". In: *German Robotics Conference (GRC)*. 2025.

** denotes equal contribution*